

Mixed effects state-space models with Student- t errors

Lina L. Hernandez-Velasco, Carlos A. Abanto-Valle & Dipak K. Dey

To cite this article: Lina L. Hernandez-Velasco, Carlos A. Abanto-Valle & Dipak K. Dey (2020) Mixed effects state-space models with Student- t errors, Journal of Statistical Computation and Simulation, 90:17, 3157-3174, DOI: [10.1080/00949655.2020.1797737](https://doi.org/10.1080/00949655.2020.1797737)

To link to this article: <https://doi.org/10.1080/00949655.2020.1797737>



Published online: 29 Jul 2020.



Submit your article to this journal [↗](#)



Article views: 170



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)



Mixed effects state-space models with Student-t errors

Lina L. Hernandez-Velasco ^a, Carlos A. Abanto-Valle^a and Dipak K. Dey^b

^aDepartment of Statistics, Federal University of Rio de Janeiro, Rio de Janeiro, Brazil; ^bDepartment of Statistics, University of Connecticut, Storrs, CT, USA

ABSTRACT

In this article, mixed-effects state space models (MESSM, [Liu D, Lu T, Niu X-F, et al. Mixed-effects state-space models for analysis of longitudinal dynamic systems. *Biometrics*. 2011;67(2):476–485.]) are revisited. MESSM can be considered as an alternative to study the HIV dynamic in a longitudinal data environment, defining the mixed-effects component into state-space models setup. As in Liu et al. [Liu D, Lu T, Niu X-F, et al. Mixed-effects state-space models for analysis of longitudinal dynamic systems. *Biometrics*. 2011;67(2):476–485.], we consider a hierarchical structure to capture possible differences between the immune systems for different patients. We extend MESSM, allowing observational errors to follow a more flexible distribution to take account for heavy tails. Using the Bayesian paradigm, an efficient Markov Chain Monte Carlo (MCMC) algorithm based on McCausland et al. [McCausland WJ, Miller S, Pelletier D. Simulation smoothing for state.space models: A computational efficiency analysis. *Comput Stat Data Anal*. 2011;55(1):199–212.] is introduced for parameter and latent variables estimation. Moreover, the mixing variables obtained as a by-product of the scale mixture representation can be used to identify outliers. The methodology is illustrated using artificial and real datasets in order to investigate the properties and performance of the proposed model.

ARTICLE HISTORY

Received 20 September 2019
Accepted 15 July 2020

KEYWORDS

State space models;
mixed-Effects; longitudinal
data; heavy tails; Bayesian
inference

1. Introduction

Human Immunodeficiency Virus (HIV) affects the immune system, specifically CD4+ T lymphocytes (called T cells), altering the DNA of these cells to make copies of itself. Once HIV multiplies, it destroys the infected cell and looks for another one to continue with the infection. HIV dynamic has been studied since the end of 1980s [1–3]. Advances in modelling this viral dynamic include to use differential equations to fit real clinical data [4,5], allowing to characterize the interaction between HIV and T cells using kinetics of HIV transmission or the clearance of the infected cells, for example. These models could be stochastic or deterministic [6]. For modelling HIV dynamic, stochastic models are recommended because, in biological phenomena, different cells and infective virus particles reacting in the same environment can often give different results [7].

CONTACT Lina L. Hernandez-Velasco  linalucia@dme.ufrj.br  Department of Statistics, Federal University of Rio de Janeiro, Caixa Postal 68530, Rio de Janeiro, CEP: 21945-970 Brazil

Usually, clinical trials designed to investigate HIV dynamic are conducted as longitudinal studies involving many individuals, collecting observations for each individual over a period of time. The main interest in these trials could be viral dynamic or the response to any treatment; both of them are described through differential/difference equations, which can be expressed into space-state models setup [8].

Liu et al. [9] proposed mixed-effect state-space models (MESSM) for longitudinal data, as an alternative to study HIV dynamic. MESSM include mixed effects components in matrices of the usual structure of state space models [10] in order to allow specific parameters for different patients, because the immune response of the subjects is not necessarily the same. MESSM results in a flexible setup to deal with existing variability, between and intra subjects. In the original specification, Liu et al. [9] impose normality assumption, for observational and state perturbations, as the basis for parameter inference. Unfortunately, normality assumption is too restrictive and suffers from the lack of robustness in the presence of outliers, which can have a significant effect on the model. To conduct Bayesian inference in MESSM context, Liu et al. [9] used single-move Gibbs sampler [11] to draw samples from posterior distribution of parameters and latent states. This sample scheme is computationally inefficient due to slow convergence.

In this article, we introduce mixed-effect state space models by considering that the observation errors follow a Student-t distribution, we named this class of models as MESSM_t . The resulting model is more flexible to accomodate outliers. Using the representation of the Student-t distribution as a scale mixture of normal distribution and the Bayesian paradigm, we develop an efficient Markov chain Monte Carlo (MCMC) algorithm for parameter estimation. It is well documented in the statistical literature that multi-move algorithms are more computationally efficient [see 12–15 for a detailed review]. Particularly, we use a multi-move procedure based on McCausland et al. [16] to draw samples for the latent states. Moreover, the mixing variables obtained as a by-product of the scale mixture representation of a Student-t distribution can be used to identify outliers.

This paper is organized as follows. In Section 2 we introduce the MESSM_t class of models and describe the proposed Bayesian inference procedure for states, mixing variables and parameters. In Section 3, a simulation study is carried out in order to verify the performance of the model and validate our inference proposal methodology. Section 4 is devoted to an application of MESSM_t to a real HIV dynamic dataset. Finally, in Section 5, we discuss possible extensions of the MESSM_t .

2. Mixed-Effect state space models with Student-t errors

2.1. Model specification

Consider a group of individuals observed in a longitudinal study. A linear MESSM_t is defined by

$$\mathbf{y}_{it} = \mathbf{G}(\boldsymbol{\theta}_i)\mathbf{x}_{it} + \mathbf{v}_{it}; \quad \mathbf{v}_{it} \sim t_{q, \nu_i}(\mathbf{0}, \mathbf{R}) \quad (1)$$

$$\mathbf{x}_{it} = \mathbf{F}(\boldsymbol{\theta}_i)\mathbf{x}_{i,t-1} + \boldsymbol{\epsilon}_{it}; \quad \boldsymbol{\epsilon}_{it} \sim \mathcal{N}_p(\mathbf{0}, \mathbf{Q}), \quad (2)$$

where $\mathbf{y}_{it}$ denote the $q \times 1$ vector of observations for the i -th individual ($i = 1, 2, \dots, m$) at time t ($t = 1, 2, \dots, n_i$), $\mathbf{x}_{it}$ is a $p \times 1$ vector of latent states, $\boldsymbol{\epsilon}_{it}$ is a $p \times 1$ dynamic disturbance vector, and $\mathbf{v}_{it}$ is a $q \times 1$ vector of observation errors. $\mathbf{R}$ and $\mathbf{Q}$ are $q \times q$ and $p \times p$

scale matrices of observations and latent states respectively, and v_i is the degree of freedom for the i -th individual. The $p \times p$ transition matrix $\mathbf{F}(\boldsymbol{\theta}_i)$ and the $q \times p$ observation-system matrix $\mathbf{G}(\boldsymbol{\theta}_i)$ are parametrized with the $r \times 1$ parameter vector $\boldsymbol{\theta}_i$, which denote all the unknown parameters in both $\mathbf{F}$ and $\mathbf{G}$, even if the parameters of these matrices are not the same. $t_{q,v}(\mu, \Sigma)$ and $\mathcal{N}_p(\mu, \Sigma)$ denote the q -variate Student-t distribution with v degrees of freedom and the p -variate normal distribution function respectively, in both cases μ denote the mean and Σ the scale matrix of the distribution. Recognizing $\boldsymbol{\theta}_i$ as the vector of individual parameters, we can model it as

$$\boldsymbol{\theta}_i = \boldsymbol{\theta} + \mathbf{b}_i, \quad \mathbf{b}_i \sim \mathcal{N}_r(\mathbf{0}, \mathbf{D}), \quad (3)$$

where $\boldsymbol{\theta}$ is the fixed effect or the population parameter vector ($r \times 1$), and the vectors $\mathbf{b}_i$ "s are the random effects, which are assumed to be i.i.d. with covariance matrix $\mathbf{D}$.

The linear MESSM_{*t*} is completely defined by Equations (1) –(3). In this formulation, $\{\mathbf{v}_{it}\}$ and $\{\boldsymbol{\epsilon}_{it}\}$ are assumed to be mutually independent within themselves and independent to each other, i.e. it is assumed that there is no relationship between the observations of a same individual over the time, or between individuals at the same time. Another assumption is that the initial values $\mathbf{x}_{i0}$'s are independent and $\mathbf{x}_{i0} \sim \mathcal{N}_p(\boldsymbol{\tau}, \mathbf{A})$. Note that the model proposed by Liu et al. [9] is a special case of a MESSM_{*t*} when every $v_i \rightarrow \infty$. However, in order to facilitate the computational performance of state estimation, we can rewrite the Equation (1) of the MESSM_{*t*} using a hierarchical representation of Student-t distribution in Scale Mixture Normal ($\mathcal{SMN}$) structure as follows,

$$\mathbf{y}_{it} = \mathbf{G}(\boldsymbol{\theta}_i)\mathbf{x}_{it} + \lambda_{it}^{-1/2}\boldsymbol{\omega}_{it}, \quad \boldsymbol{\omega}_{it} \sim \mathcal{N}_q(\mathbf{0}, \mathbf{R}), \quad (4)$$

$$\lambda_{it} \sim \mathcal{G}\left(\frac{v_i}{2}, \frac{v_i}{2}\right), \quad (5)$$

where $\mathcal{G}(\cdot, \cdot)$ denotes gamma distribution. So, we are going to work with the MESSM_{*t*} defined by Equations (2), (3) and (4).

2.2. Bayesian inference via MCMC

Let $\boldsymbol{\varphi} = (\mathbf{R}, \mathbf{Q}, \mathbf{D}, \boldsymbol{\theta}, \Theta, \mathbf{v})^T$ be the full parameter vector of MESSM_{*t*}, where $\mathbf{R}, \mathbf{Q}, \mathbf{D}$ and $\boldsymbol{\theta}$ were previously defined. Θ is a $r \times m$ matrix containing the $\boldsymbol{\theta}_i$'s vectors of individual parameters and $\mathbf{v} = (v_1, \dots, v_m)^T$ is the vector of degrees of freedom. We also consider $\boldsymbol{\lambda} = (\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_m)$ where $\boldsymbol{\lambda}_i$ is a $1 \times n_i$ vector of mixing variables associated with each patient. Let $\mathbf{x}_i = (\mathbf{x}_{i1}, \dots, \mathbf{x}_{in_i})$ be the state variables matrix and $\mathbf{y}_i = (\mathbf{y}_{i1}, \dots, \mathbf{y}_{in_i})$ be the matrix of observations for each individual respectively. By using Bayes' theorem, the joint posterior density of parameters and state variables can be written as:

$$p(\boldsymbol{\varphi}, \boldsymbol{\lambda}, \mathbf{x}_1, \dots, \mathbf{x}_m | \mathbf{y}_1, \dots, \mathbf{y}_m) \propto p(\boldsymbol{\varphi}) \prod_{i=1}^m p(\boldsymbol{\lambda}_i) p(\mathbf{y}_i | \mathbf{x}_i, \boldsymbol{\varphi}, \boldsymbol{\lambda}_i) p(\mathbf{x}_i | \boldsymbol{\varphi}), \quad (6)$$

where

$$p(\mathbf{y}_i | \mathbf{x}_i, \boldsymbol{\varphi}, \boldsymbol{\lambda}) \propto \prod_{t=1}^{n_i} \exp \left\{ -\frac{\lambda_{it}}{2} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i)\mathbf{x}_{it})^T \mathbf{R}^{-1} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i)\mathbf{x}_{it}) \right\}$$

$$\begin{aligned}
p(\mathbf{x}_i|\boldsymbol{\varphi}) &\propto \prod_{t=1}^{n_i} \exp \left\{ -\frac{1}{2} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i)\mathbf{x}_{i,t-1})^T \mathbf{Q}^{-1} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i)\mathbf{x}_{i,t-1}) \right\} \\
p(\boldsymbol{\varphi}) &\propto p(\mathbf{Q})p(\mathbf{R})p(\mathbf{D})p(\boldsymbol{\theta}) \prod_{i=1}^m \exp \left\{ -\frac{1}{2} (\boldsymbol{\theta}_i - \boldsymbol{\theta})^T \mathbf{D}^{-1} (\boldsymbol{\theta}_i - \boldsymbol{\theta}) \right\} \\
p(\lambda_{it}) &\propto \lambda_{it}^{(v_i/2)-1} \exp \left\{ -\frac{v_i}{2} \lambda_{it} \right\}.
\end{aligned} \tag{7}$$

Since the posterior density $p(\boldsymbol{\varphi}, \mathbf{x}_1, \dots, \mathbf{x}_m, \boldsymbol{\lambda}|\mathbf{y}_1, \dots, \mathbf{y}_m)$ does not have closed form, first we sample the parameters and then the state variables using a Gibbs sampling scheme described by Algorithm 2.1

- Algorithm 2.1:** 1) Set $j = 0$ and get starting values for the parameters $(\mathbf{R}^{(0)}, \mathbf{Q}^{(0)}, \mathbf{D}^{(0)}, \boldsymbol{\theta}^{(0)}, \mathbf{v}^{(0)})$ and the mixing variables $\boldsymbol{\lambda}^{(0)}$. **For** $j \geq 1$
- 2) For each individual i in $\{1, 2, \dots, m\}$, generate
 - $\mathbf{x}_i^{(j)}$ using Algorithm 2.2.
 - $\boldsymbol{\theta}_i^{(j)} \sim p(\boldsymbol{\theta}_i|\boldsymbol{\theta}^{(j)}, \mathbf{x}_i^{(j)}, \mathbf{Q}^{(j)}, \mathbf{D}^{(j)})$.
 - $\lambda_{it}^{(j)} = (\lambda_{i1}^{(j)}, \lambda_{i2}^{(j)}, \dots, \lambda_{in}^{(j)})$ using $\lambda_{it}^{(j)} \sim p(\lambda_{it}|\mathbf{x}_{it}^{(j-1)}, \mathbf{y}_{it}, \mathbf{R}^{(j)}, v_i^{(j-1)})$ for each $t = 1, 2, \dots, n_i$.
 - $v_i^{(j)} \sim p(v_i|\lambda_i^{(j)})$.
 - 3) Generate
 - $\mathbf{R}^{(j)} \sim p(\mathbf{R}^{-1}|\mathbf{x}^{(j)}, \mathbf{y}, \boldsymbol{\Upsilon}_2^{-1}, \rho_2)$
 - $\boldsymbol{\theta}^{(j)} \sim p(\boldsymbol{\theta}|\boldsymbol{\Theta}^{(j)}, \boldsymbol{\eta}, \mathbf{D}^{(j-1)}, \boldsymbol{\Delta})$
 - $\mathbf{Q}^{(j)} \sim p(\mathbf{Q}^{-1}|\mathbf{x}^{(j)}, \boldsymbol{\Theta}, \boldsymbol{\Upsilon}_1, \rho_1)$
 - $\mathbf{D}^{(j)} \sim p(\mathbf{D}^{-1}|\boldsymbol{\Theta}^{(j)}, \boldsymbol{\theta}^{(j)}, \boldsymbol{\Upsilon}_3^{-1}, \rho_3)$

where $p(\boldsymbol{\theta}|\boldsymbol{\Theta}, \boldsymbol{\eta}, \mathbf{D}, \boldsymbol{\Delta})$, $p(\mathbf{Q}^{-1}|\mathbf{x}, \boldsymbol{\Theta}, \boldsymbol{\Upsilon}_1, \rho_1)$, and $p(\mathbf{D}^{-1}|\boldsymbol{\Theta}, \boldsymbol{\theta}, \boldsymbol{\Upsilon}_3^{-1}, \rho_3)$ are the full conditional distribution of $\boldsymbol{\theta}$, $\mathbf{Q}^{-1}$, and $\mathbf{D}^{-1}$. Their analytical expressions are given by Equations (A1), (A2) and (A3) respectively, in [appendix 1](#).
 - 4) Set $j = j + 1$ and return to 2 until convergence is achieved.

2.2.1. Sampling parameters and mixing variables

Here we must consider that there are two types of parameters within $\boldsymbol{\varphi}$ set; the individual parameters $\boldsymbol{\varphi}_I$, that correspond to each $\boldsymbol{\theta}_i \in \boldsymbol{\Theta}$ and $v_i \in \mathbf{v}$; and $\boldsymbol{\varphi}_r$ which correspond to the rest of the parameters that does not depend on the individual. Particularly, $\boldsymbol{\theta}$ is the *population parameter*. For the $\boldsymbol{\varphi}_r$ parameters $\mathbf{R}, \mathbf{Q}, \mathbf{D}$ and $\boldsymbol{\theta}$, we use conjugate priors that usually leave to a closed form for the posterior distribution, allowing us to use the Gibbs sampler similarly as done by Liu et al. [9]. We set the priors as follows: $\mathbf{Q}^{-1} \sim W(\boldsymbol{\Upsilon}_1^{-1}, \rho_1)$, $\mathbf{R}^{-1} \sim W(\boldsymbol{\Upsilon}_2^{-1}, \rho_2)$, $\mathbf{D}^{-1} \sim W(\boldsymbol{\Upsilon}_3^{-1}, \rho_3)$ and $\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\eta}, \boldsymbol{\Delta})$, where $W(\boldsymbol{\Upsilon}, \rho)$ denotes the Wishart distribution with matrix scale $\boldsymbol{\Upsilon}$ and degrees of freedom ρ . Note that if matrices $\mathbf{Q}, \mathbf{R}$ and $\mathbf{D}$ are such that $\mathbf{Q} = \sigma_Q^2 \mathbf{I}$, $\mathbf{R} = \sigma_R^2 \mathbf{I}$ and $\mathbf{D} = \sigma_D^2 \mathbf{I}$, we can use gamma priors for σ_Q^{-2} , σ_R^{-2} and σ_D^{-2} respectively.

For $\boldsymbol{\varphi}_I$ parameters, we first consider the following full posterior distribution for each $\boldsymbol{\theta}_i \in \boldsymbol{\Theta}$ as consequence of Equations (2) and (3)

$$\boldsymbol{\theta}_i | \boldsymbol{\theta}, \mathbf{x}, \mathbf{Q}, \mathbf{D} \sim \mathcal{N}(\mathbf{V}_2 \mathbf{b}_2, \mathbf{V}_2), \quad (8)$$

where $\mathbf{V}_2$ and $\mathbf{b}_2$ are described in appendix 1.

For each degree of freedom v_i , we considered the Jeffrey's prior Fonseca et al. [17] that leads to the full conditional

$$p(v_i | \lambda_{it}) \propto p(v_i) \left(\frac{v_i}{2}\right)^{(v_i n_i / 2)} \left[\Gamma\left(\frac{v_i}{2}\right)\right]^{-n_i} \exp \left\{ -v_i \left[\frac{1}{2} \sum_{t=1}^{n_i} (\log(\lambda_{it}) + \lambda_{it}) \right] \right\}. \quad (9)$$

For mixing variables λ_{it} the full conditional is given by

$$\lambda_{it} | \mathbf{x}_{it}, \mathbf{y}_{it}, \mathbf{R}, v_i \sim \mathcal{G}(\alpha_i, \beta_i), \quad (10)$$

where $\alpha_i = ((v_i + 1)/2)$ and $\beta_i = \frac{1}{2}(\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i)\mathbf{x}_{it})^T \mathbf{R}^{-1}(\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i)\mathbf{x}_{it}) + v_i$ for $i = 1, \dots, m$ and $t = 1, \dots, n_i$. Given that Equation (10) has closed form, we sampled directly from it.

2.2.2. Sampling states

McCausland et al. [16] proposed the McCausland-Miller-Pelletier (MMP) method which is based on recognizing that the conditional distribution of states given observation in Gaussian linear state-space models is a special case of a Gaussian Markov Random Fields (GMRF). Our interest is to draw $\mathbf{x}_i$ from $\mathbf{x}_i | \mathbf{y}_i \sim \mathcal{N}(\boldsymbol{\Omega}^{-1} \mathbf{c}, \boldsymbol{\Omega}^{-1})$. Applying the MMP method, we used the following algorithm to draw the states for every individual:

Algorithm 2.2: (1): Pre-computation for $t = 1, 2, \dots, n$,

- Compute the Cholesky decomposition $\boldsymbol{\Sigma}_t^{-1} = \boldsymbol{\Lambda}_t \boldsymbol{\Lambda}_t^T$, where $\boldsymbol{\Sigma}_1^{-1} = \boldsymbol{\Omega}_{11}$ and $\boldsymbol{\Sigma}_t^{-1} = \boldsymbol{\Omega}_{tt} - [\boldsymbol{\Omega}_{t-1,t}^T \boldsymbol{\Sigma}_{t-1}^{-1} \boldsymbol{\Omega}_{t-1,t}]$ for $t > 1$.
- Compute $\boldsymbol{\Lambda}_t^{-1} \boldsymbol{\Omega}_{t,t+1}$ using triangular back-substitution.
- Compute the symmetric matrix $\boldsymbol{\Omega}_{t,t+1}^T \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\Omega}_{t,t+1} = [\boldsymbol{\Lambda}_t^{-1} \boldsymbol{\Omega}_{t,t+1}]^T [\boldsymbol{\Lambda}_t^{-1} \boldsymbol{\Omega}_{t,t+1}]$.

(2): Drawing $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\Omega}^{-1} \mathbf{c}, \boldsymbol{\Omega}^{-1})$ in backwards. For $t = n, n-1, \dots, 1$,

- Draw $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(0, \mathbf{I}_p)$.
- Compute $\mathbf{x}_t$ using matrix-vector multiplication and back-substitution, where $\mathbf{x}_n = \mathbf{m}_n + (\boldsymbol{\Lambda}_n)^{-1} \boldsymbol{\varepsilon}_n$ and $\mathbf{x}_t = \mathbf{m}_t + (\boldsymbol{\Lambda}_t)^{-1} (\boldsymbol{\varepsilon}_t - [\boldsymbol{\Lambda}_t^{-1} \boldsymbol{\Omega}_{t,t+1}] \mathbf{x}_{t+1})$ for $t < n$.

In Algorithm 2.2, we assume that the full parameter vector $\boldsymbol{\varphi} = (\boldsymbol{\varphi}_r, \boldsymbol{\varphi}_I)^T$ is known. Details for this algorithm applied to MESSM_t are given in appendix 2.

3. Simulation

With the objective to evaluate the computational efficiency and empirical properties of the estimators obtained using the proposed sampling schemes, we carried out two simulation experiments. First, in order to compare the performance of the single-move and the proposed multi-move algorithm, we simulate a single dataset from univariate model as in [9]. Secondly, we conduct a Monte Carlo study from a bivariate model using it to show frequentist properties of the estimators in MCMC simulation.

3.1. Comparison in states estimation

We consider a MESSM_t model for $i = 1, \dots, 50$ and $t = 1, \dots, 20$ for each individual i defined by

$$\begin{aligned} y_{it} &= x_{it} + \omega_{it}; & \omega_{it} | \lambda_{it} &\sim \mathcal{N}(0, \lambda_{it}^{-1} \sigma_R^2), & \lambda_{it} &\sim \mathcal{G}\left(\frac{\nu_i}{2}, \frac{\nu_i}{2}\right) \\ x_{it} &= \theta_i x_{i,t-1} + \epsilon_{it}; & \epsilon_{it} &\sim \mathcal{N}(0, \sigma_Q^2) \\ \theta_i &= \theta + b_i; & b_i &\sim \mathcal{N}(0, \sigma_D^2), \end{aligned}$$

where $\sigma_R^2 = 1$, $\sigma_Q^2 = 1.5$, $\sigma_D^2 = 0.005$, $\nu_i = \nu = 11$ and population parameter $\theta = 0.85$. We ran 10000 Gibbs iterations discarding the first 5000 for each scheme, single-move (*Single*) and multi-move (*MMP*). For comparison, we calculate the inefficiency factor (IF) defined by $1 + \sum_{s=1}^{\infty} \rho_s$ where ρ_s is the sample autocorrelation at lag s . The IF measures how well the MCMC chain mixes [18] and it is estimated as the ratio between the numerical variance of the posterior sample mean and the variance of the sample mean from uncorrelated draws. When the inefficiency factor is equal to k , we need to draw MCMC samples k times as many as the number of uncorrelated samples.

In Figures 1 and 2, we show the autocorrelation function (ACF) of four selected states of an individual and some parameters obtained from the MCMC output. They show that the draws using the MMP sampler are less correlated. Table 1 shows the IF for the chains of an individual state in four different times ($t = 1, 5, 10$ and 20), and parameters θ , σ_R^2 and σ_Q^2 . Comparing the IF of the methods in each case, we found that this is always lower for *MMP* and then this method produce better mixing of the MCMC chains.

3.2. A bivariate MESSM_t example

Now, consider a MESSM_t defined by

$$y_{it} = \mathbf{G}\mathbf{x}_{it} + \omega_{it}; \quad \omega_{it} | \lambda_{it} \sim \mathcal{N}(0, \lambda_{it}^{-1} \sigma_R^2), \quad \lambda_{it} \sim \mathcal{G}\left(\frac{\nu_i}{2}, \frac{\nu_i}{2}\right)$$

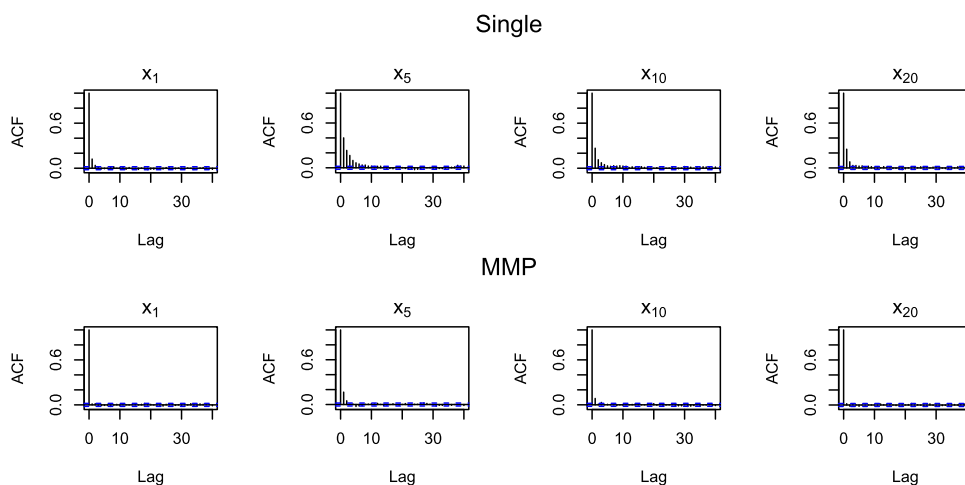


Figure 1. ACF for states x_1, x_5, x_{10} and x_{20} of an individual.

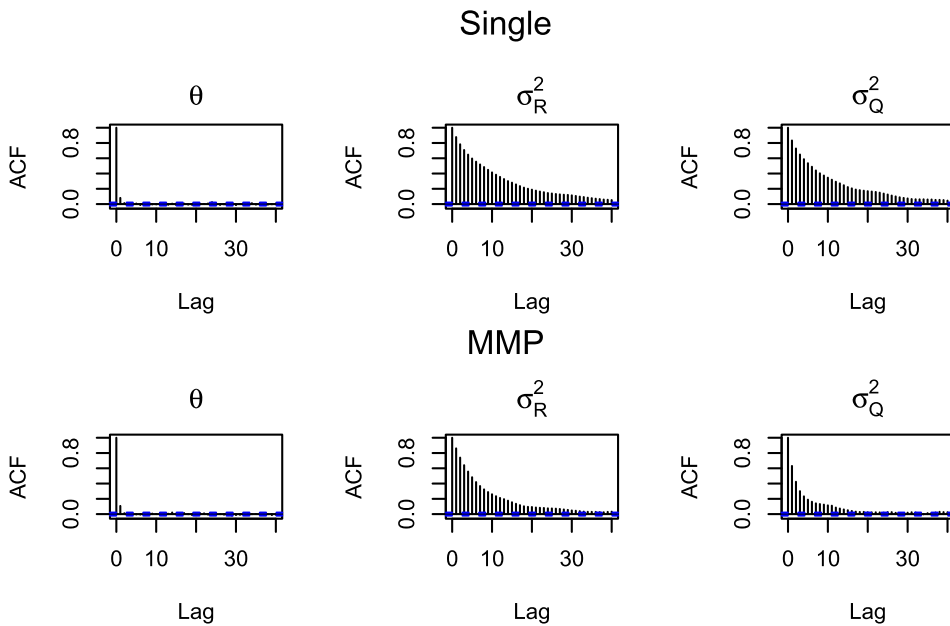


Figure 2. ACF for parameters θ , σ_R^2 and σ_Q^2 .

Table 1. Inefficiency Factor.

Chain	x_1	x_5	x_{10}	x_{20}	θ	σ_R^2	σ_Q^2
<i>Single</i>	1.45	1.66	1.63	2.03	1.27	23.09	19.69
<i>MMP</i>	1.19	1.1	1.04	1.21	1.24	13.76	7.23

$$\mathbf{x}_{it} = \mathbf{F}(\boldsymbol{\theta}_i)\mathbf{x}_{i,t-1} + \boldsymbol{\epsilon}_{it}; \quad \boldsymbol{\epsilon}_{it} \sim \mathcal{N}_2(\mathbf{0}, \sigma_Q^2 \mathbf{I}_2)$$

$$\boldsymbol{\theta}_i = \boldsymbol{\theta} + \mathbf{b}_i, \quad \mathbf{b}_i \sim \mathcal{N}_2(\mathbf{0}, \sigma_D^2 \mathbf{I}_2),$$

where $\mathbf{G} = \begin{pmatrix} 1 & 1 \end{pmatrix}$, $\sigma_R^2 = 1$, $\boldsymbol{\theta} = (0.3, 0.6)^T$, $\sigma_D^2 = 0.0064$, $\sigma_Q^2 = 1.44$, $\mathbf{F}_i = \begin{pmatrix} 0.7 & 0 \\ \theta_{1i} & \theta_{2i} \end{pmatrix}$ and $v_i = 11$ for every $i = 1, \dots, m$.

We assume $\sigma_D^{-2} \sim \mathcal{G}(0.5; 0.001)$, $\sigma_Q^{-2} \sim \mathcal{G}(0.5; 0.5)$, $\sigma_R^{-2} \sim \mathcal{G}(0.5; 0.5)$ and $\boldsymbol{\theta} \sim \mathcal{N}_2((0.5, 0.5)^T, 2\mathbf{I}_2)$ as priors, where the hyperparameter was chosen as in Liu et al. [9]. We accomplish a simulation study with 50 replicates for $m = 20$ and $n_i = n = 20$, running 20000 Gibbs iterations discarding the first 5000 draws as burn-in for each replicate. The estimation results are shown in Table 2 where *MSE* is the mean square error, *MSRE* is the relative error of the square root of *MSE* computed as $\sqrt{\text{MSE}}/(\text{True Parameter})$ and Coverage percentage (CP) represents the percentage of times that the unknown quantity is inside the credibility interval 95%. From Table 2, we conclude that the estimates of θ_1 and θ_2 were very close to the true values, but the variability of θ_2 was smaller than for θ_1 . With respect to the *MSRE* we can say that the most accurate estimation was $\hat{\theta}_2$ while the worst was $\hat{\sigma}_D^2$, the last one can be explain because the true value of this parameter is too small. The coverage of all parameters is more than 90% which is reasonably good.

Table 2. Parameter estimation for the bivariate MESSM_t model with 50 replicates, $m = 20$ and $n = 20$.

	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\sigma}_D^2$	$\hat{\sigma}_Q^2$	$\hat{\sigma}_R^2$	$\hat{\nu}_6$
True	(0.3)	(0.6)	(0.0064)	(1.44)	(1)	(11)
Mean	0.3042	0.5930	0.0056	1.5959	0.8333	12.1125
Bias	0.0042	-0.0070	-0.000084	0.1559	-0.1667	1.1125
MSE	0.0018	0.0016	0.000009	0.0399	0.0329	1.4456
MSRE	14.05%	6.67%	48.30%	17.60%	24.64%	14.89%
CP	98%	98%	92%	98%	92%	100%

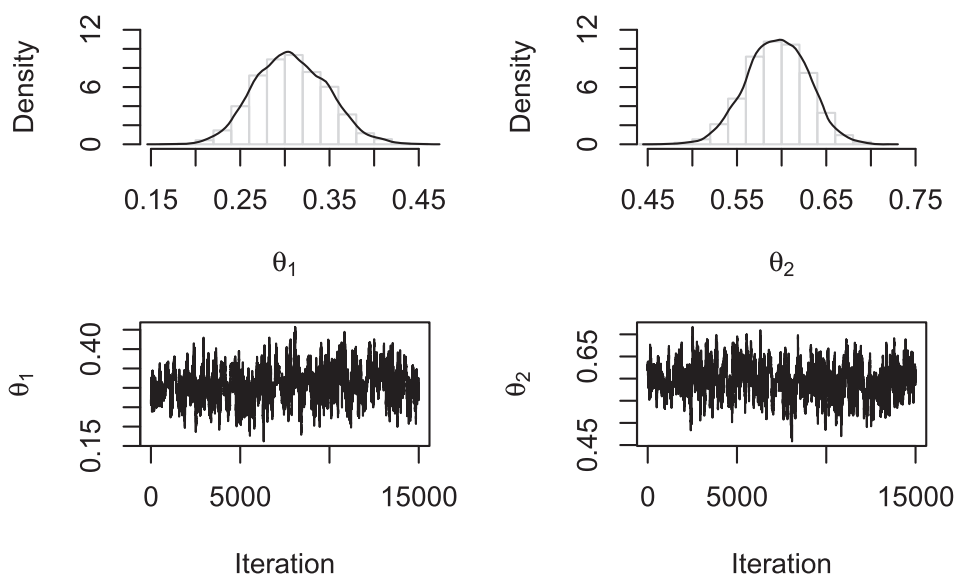
**Figure 3.** Parameter estimation results of θ_1 and θ_2 for MESSM_t .

Figure 3 shows the histogram of θ_1 and θ_2 with the respectively density estimation for a single replicate. The histogram shows that the draws are centred around the true values (0.3 for θ_1 and 0.6 for θ_2). Figure 3 also show the trace plots of these parameters, which suggest the convergence to a stationary distribution. Also we found that trace plots for the other parameters (not shown) suggest the convergence of the Gibbs samplers.

4. Application in real data

In this section, we apply the methods described in Section 2 to a real dataset from the AIDS Clinical Trial Group (ACTG) Study 315 [19]. In this trial, 48 HIV-infected patients were treated with potent antiviral drugs (ritonavir, 3TC and AZT); Treatments started on day 0 and the plasma concentration of HIV-1 RNA was repeatedly measured on days 0, 2, 7, 10, 14, 21, 28 and weeks 8, 12 after the initiation of the treatment. However, for multiple reasons, the days in which each patient was observed do not coincide exactly with the days mentioned before and some of them have less information than others. For example, patients 2, 29 and 33 were observed less than 3 times. This does not constitute a problem for the model itself, since it considers that each patient can be observed a different number

of days. In their application, Liu et al. [9] used a two-phase clearance of viral load because, as suggested by Perelson et al. [4], the first phase featured a rapid drop in viral load due to the exponential death rate of the productively infected cells during the first four weeks of treatment. Liu et al. [9] assumed that rate of clearance changes at the mid-point $t = 24$ and fitted the model considering $\theta_{it} = \theta_i^{(1)}$ for $t \leq 23$ and $\theta_{it} = \theta_i^{(2)}$ for $t > 23$. Due to the lack information available for both phases, we use the simple HIV dynamic model with a constant rate based on [20]

$$\begin{aligned} y_{i,t} &= x_{i,t} + \omega_{i,t}, & \omega_{i,t} | \lambda_{i,t} &\sim \mathcal{N}(0, \lambda^{-1} \sigma_R^2), & \lambda_{i,t} &\sim \mathcal{G}\left(\frac{\nu_i}{2}, \frac{\nu_i}{2}\right) \\ x_{i,t} &= \theta_i x_{i,t-1} + \epsilon_{i,t}, & \epsilon_{i,t} &\sim \mathcal{N}(0, \sigma_Q^2) \\ \theta_i &= \theta + b_i, & b_i &\sim \mathcal{N}(0, \sigma_D^2) \end{aligned} \quad (11)$$

where $y_{i,t}$ is the base 10 logarithm of the measured viral load from patient i at time t , $x_{i,t}$ represents the base 10 logarithm of the actual viral load and θ_i indicates the viral clearance rate for patient i . For this application, we set $\sigma_D^{-2} \sim \mathcal{G}(0.01; 0.01)$, $\sigma_Q^{-2} \sim \mathcal{G}(0.01; 0.01)$, $\sigma_R^{-2} \sim \mathcal{G}(0.01; 0.01)$ and $\theta \sim \mathcal{N}(0.5, 10)$ as prior distributions. We fit a MESSM_t, according with model in (11), running 10000 Gibbs iterations discarding the first 5000 draws for estimation. Similarly, in order to compare the results, we fit the corresponding MESSM.

Note that the observations were taken from specific days in a period of approximately four months. This means that we have a maximum of eight observations for each patient, and the rest of days should be considered as missing. Unfortunately, The original MMP was formulated for equally spaced observations, so it cannot applied directly in presence of missing values. In order to overcome this inconvenient, we create an artificial path of observation **ymiss_i**, for each individual i . We assume that missing values follow the same data process generator for y_{it} in Equation (11), and we use it to add a Gibbs sampler step to draw y_{it} when it is missing. In each step of MCMC, we use $\mathbf{y}_i = (\mathbf{yobs}_i, \mathbf{ymiss}_i)^T$ as the observation vector, when **yobs_i** denote the vector of truly observed values.

In Table (3), we report the posterior mean and the 95% credibility intervals (CI) of the parameters θ , σ_R^2 , σ_Q^2 and σ_D^2 . Estimation results are similar for both fittings for almost every parameter, except for the scale parameter of the observations (σ_R^2). Values from point estimation of σ_Q^2 are shown different, but the CI 95% intersect, so we do not reject the possibility that the value of this parameter is similar in both models. The clearance rate value (θ) is about 0.985 for both models and it means that the viral load is declines in at least 1.5% per day.

In order to compare the performances of the models MESSM and MESSM_t, we use the deviance information criterion (DIC), proposed by Spiegelhalter et al. [21]:

$$\text{DIC} = -2 \sum_{i=1}^m \sum_{t=1}^{n_i} \log(p(y_{it} | \sigma_R^2, \nu_i, \lambda_{it})) - p_{\text{DIC}},$$

where p_{DIC} measures the complexity of the model by the effective number of parameters, and σ_R^2 , ν_i and λ_{it} are the posterior mean of parameters and mixing variables on which conditional distribution of y_{it} depends. Define the deviance of the model as $D(\boldsymbol{\varphi}) = -2 \sum_{i=1}^m \sum_{t=1}^{n_i} \log(p(y_{it} | \boldsymbol{\varphi}))$. p_{DIC} is defined as the difference between the posterior mean of the deviance and the deviance evaluated at the posterior mean of $\hat{\sigma}_R^2$, $\hat{\nu}_i$ and $\hat{\lambda}_{it}$. The

DIC can be estimated using the MCMC outputs ($s = 1, \dots, S$) by $\widehat{DIC} = 2\bar{D} - \widehat{D}$, where $\bar{D} = \sum_{s=1}^S D(\boldsymbol{\varphi}_s)/S$, $\widehat{D} = D(\widehat{\boldsymbol{\varphi}})$ and $\widehat{\boldsymbol{\varphi}}$ is the posterior expected value of $\boldsymbol{\varphi} = (\sigma_R^2, \nu_i, \lambda_{it})$. However, DIC suffers from some theoretical aspects [see 22–24 for more details]. As an alternative, we also use the Watanabe Akaike information criterion (W_{AIC}), proposed by Watanabe [25]. Compared to DIC, W_{AIC} has the desirable property of averaging over the posterior distribution rather than conditioning on a point estimate. This is especially relevant in a predictive context, as W_{AIC} is evaluating the predictions that are actually being

Table 3. Parameter estimation of viral load for HIV dynamic model (11) using MESSM and MESSM_t.

Parameter	MESSM		MESSM _t	
	Mean	CI(95%)	Mean	CI(95%)
θ	0.9848	[0.9771;0.9919]	0.9847	[0.9772;0.9921]
σ_R^2	0.2369	[0.1775;0.2989]	0.068	[0.0349;0.1031]
σ_Q^2	0.0117	[0.0037;0.0219]	0.0170	[0.0072;0.0286]
σ_D^2	0.000623	[0.00040;0.00099]	0.000627	[0.00039;0.00098]
DIC	1.45		1.27	
WAIC	433.43		370.01	

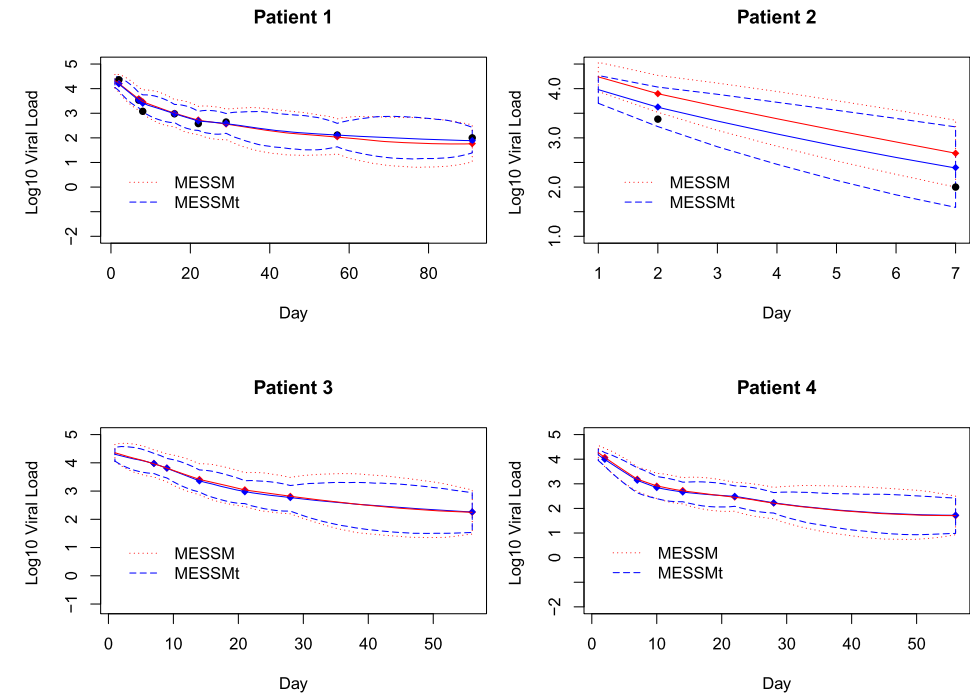


Figure 4. Estimation and prediction of viral load for four patients in the HIV dynamic study. Base 10 logarithm of the viral loads are shown in black circles. The smoothing estimates of states are shown in solid curves, the 95% error bounds are shown by dashed curves for MESSM_t and dotted curves for MESSM. The prediction of each observation are shown by color points, red represents the normal model (MESSM) and blue represents the Student-t model (MESSM_t).

used for new data in a Bayesian context [23]. W_{AIC} is computed as

$$\widehat{W}_{AIC} = \sum_{i=1}^m \sum_{t=1}^{n_i} \log \left(\sum_{s=1}^S p(y_{it} | \boldsymbol{\varphi}_s) \right) - 2 \sum_{i=1}^m \sum_{t=1}^{n_i} \left(\log \left(\frac{1}{S} \sum_{s=1}^S p(y_{it} | \boldsymbol{\varphi}_s) \right) - \frac{1}{S} \sum_{s=1}^S \log p(y_{it} | \boldsymbol{\varphi}_s) \right).$$

In both cases, the model with smallest DIC and W_{AIC} has best fit. According to Table 3, the DIC and W_{AIC} values indicate that the MESSM_t has the best fit (see values in bold).

Figure 4 gives the results for the first four patients. We can say for patients 1, 3 and 4 that the estimations of both models are very similar and closer to the observations but for MESSM_t approach we obtain a little reduction in the 95% error bounds for the states when observations are available. For patient 2, we found that MESSM_t shows better fit because just for this model the observed viral loads (black circles) are include inside the 95% error bounds.

In the Bayesian paradigm, the posterior mean of the mixing variables can be used to identify a possible outlier [see 26]. The MESSM_t can accommodate an outlier by inflating the variance component for that observation in the conditional normal distribution with λ_{it} value smaller than 1. In Figure 5, black points represent the small values of $\hat{\lambda}_{it}$ for the first four patients. We identify these points as the possible outliers for each patient.

Table 4 shows that degrees of freedom for some patients are small. So, we conclude that there is evidence that the distribution associated to the observational disturbance

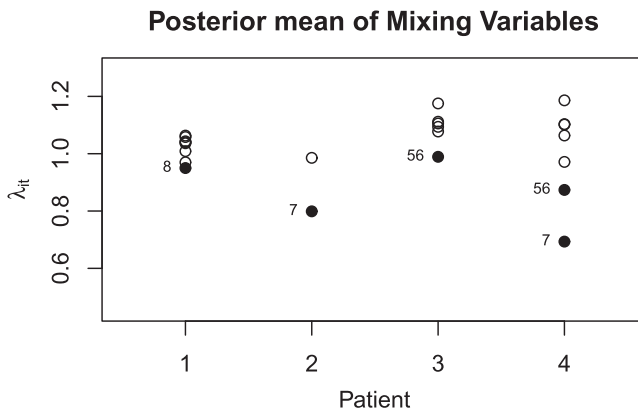


Figure 5. Posterior mean of mixing variables λ_{it} for the first four patients (white points). The number represents the day t of the observation. Black points are the possible outliers.

Table 4. Degree of freedom estimations for HIV dynamic model (11) using MESSM_t .

	$\hat{\nu}_1$	$\hat{\nu}_2$	$\hat{\nu}_3$	$\hat{\nu}_4$
Mean	13.62	4.03	4.06	4.07
CI(95%)	[4.29; 35.50]	[4.00; 4.10]	[4.00; 4.23]	[4.00; 4.24]

has heavy tails. This confirms what we wrapped up previously based on the comparison criteria.

5. Discussions

In this paper, we introduced the mixed-effects state-space model with Student- t errors as an extension of MESSM proposed by Liu et al. [9]. Our proposed method allows to the observational disturbance to follow a more flexible distribution to take account for outliers. In particular, we consider a Student- t distribution for the observational error, introducing a hierarchical structure on the observational equation of the space-state form in order to simplify the computational performance of the state estimation. Under a Bayesian perspective, we develop an efficient MCMC sampling procedure to estimate all the parameters and latent quantities in our proposed MESSM $_t$. Through simulation studies, we showed empirically that using a multi-move algorithm (as MMP) to generate samples of the states, we improve the MCMC performance obtaining better mixing properties. We also conclude that our methodology has a well performance even in more complicated cases as considering bivariate states. With the intention of giving a comprehensive analysis of our model, we used an application in real data to illustrate the performance of our extension comparative with MESSM. We found that MESSM $_t$ was more flexible modelling the observations, showing better fitting properties than MESSM.

We have extended the MESSM to a more flexible class of models that we called MESSM $_t$. This new approach considers the potential difference among responses in longitudinal dynamic systems, which allows to deal with more applications in longitudinal data analysis. The original MESSM proposed by Liu et al. [9] had three assumptions: normality, linearity and no correlation. With our proposal we have made flexible the first assumption allowing the use of a different distribution which better fits outliers. However, our model continues restricting its applications just for the linear case, that is quite limited in the HIV dynamic context. Another extensions can be investigated. For example, [27] extended the MESSM to a non-linear case and [28] proposed a generalized MESSM which can accommodate the groups with correlated individuals. In clinical trials context, individuals used to be uncorrelated and then the generalization of [28] does not apply to these cases. We did not compare MESSM $_t$ against the proposal of [27], because we are in disadvantage. Our proposal is restricted to the linear case, therefore the comparison would not be fair. It is known that a non linear approach is better to fit HIV dynamic [3], for this reason, we underline the importance of studying the possibility of extending the MESSM $_t$ for the non-linear case. Also, it would be interesting to adapt the MMP sampler to be used directly to the dataset with missing values and comparing the results with those obtained using the artificial path. This is important because the majority of the HIV dynamic datasets have missing observations.

Acknowledgements

The authors would like to thank the Editor and an anonumour referee for their useful comments, which improved the quality of this article. The first author gratefully acknowledges financial support from the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES). The research of Carlos A. Abanto-Valle was supported in part by the Fundação de Amparo à Pesquisa do Estado de Rio de Janeiro (FAPERJ).

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Lina L. Hernandez-Velasco  <http://orcid.org/0000-0001-7931-793X>

References

- [1] May RM, Anderson RM. Transmission dynamics of hiv infection. *Nature*. 1987;326(6109):137–142.
- [2] Perelson AS, Kirschner DE, De Boer R. Dynamics of hiv infection of cd4+ t cells. *Math Biosci*. 1993;114(1):81–125.
- [3] Perelson AS, Nelson PW. Mathematical analysis of hiv-1 dynamics in vivo. *SIAM Rev*. 1999;41(1):3–44.
- [4] Perelson AS, Essunger P, Cao Y, et al. Decay characteristics of hiv-1-infected compartments during combination therapy. *Nature*. 1997;387(6629):188–191.
- [5] Wei X, Ghosh SK, Taylor ME, et al. Viral dynamics in human immunodeficiency virus type 1 infection. *Nature*. 1995;373(6510):117–122.
- [6] Tan W-Y, Wu H. Deterministic and stochastic models of AIDS epidemics and HIV infections with intervention. London: World Scientific Publishing Co. Pte. Ltd; 2005.
- [7] Dalal N, Greenhalgh D, Mao X. A stochastic model for internal hiv dynamics. *J Math Anal Appl*. 2008;341(2):1084–1101.
- [8] Tan W-Y, Zhang P, Xiong X. A state space model for HIV pathogenesis under anti-viral drugs and application. In: Tan W-Y, Wu H, editors. Deterministic and stochastic models of AIDS epidemics and HIV infections with intervention. London: World Scientific Publishing Co. Pte. Ltd; 2005. p. 333–359.
- [9] Liu D, Lu T, Niu X-F, et al. Mixed-effects state-space models for analysis of longitudinal dynamic systems. *Biometrics*. 2011;67(2):476–485.
- [10] West M, Harrison J. Bayesian forecasting and dynamic models. New York (NY): Springer-Verlag; 1997.
- [11] Carlin BP, Polson NG, Stoffer DS. A monte carlo approach to nonnormal and nonlinear state-space modeling. *J Am Stat Assoc*. 1992;87(418):493–500.
- [12] Carter CK, Kohn R. On gibbs sampling for state space models. *Biometrika*. 1994;81(3):541–553.
- [13] De Jong P, Shephard N. The simulation smoother for time series models. *Biometrika*. 1995;82(2):339–350.
- [14] Durbin J, Koopman SJ. A simple and efficient simulation smoother for state space time series analysis. *Biometrika*. 2002;89(3):603–616.
- [15] Frühwirth-Schnatter S. Data augmentation and dynamic linear models. *J Time Ser Anal*. 1994;15(2):183–202.
- [16] McCausland WJ, Miller S, Pelletier D. Simulation smoothing for state-space models: A computational efficiency analysis. *Comput Stat Data Anal*. 2011;55(1):199–212.
- [17] Fonseca TC, Ferreira MA, Migon HS. Objective bayesian analysis for the student-t regression model. *Biometrika*. 2008;95(2):325–333.
- [18] Kim S, Shephard N, Chib S. Stochastic volatility: likelihood inference and comparison with arch models. *Rev Econ Stud*. 1998;65(3):361–393.
- [19] Wu H, Ding AA. Population hiv-1 dynamics in vivo: applicable models and inferential tools for virological data from aids clinical trials. *Biometrics*. 1999;55(2):410–418.
- [20] Ho DD, Neumann AU, Perelson AS, et al. Rapid turnover of plasma virions and cd4 lymphocytes in hiv-1 infection. *Nature*. 1995;373(6510):123–126.
- [21] Spiegelhalter DJ, Best NG, Carlin BP, et al. Bayesian measures of model complexity and fit. *J R Stat Soc Ser B (Stat Methodol)*. 2002;64(4):583–639.

- [22] Celeux G, Forbes F, Robert CP, et al. Deviance information criteria for missing data models. *Bayesian Anal.* **2006**;1(4):651–673.
- [23] Gelman A, Hwang J, Vehtari A. Understanding predictive information criteria for bayesian models. *Stat Comput.* **2014**;24(6):997–1016.
- [24] Stone M. ‘Discussion on “bayesian measures of model complexity and fit”’. *J R Stat Soc Ser B.* **2002**;0:00–00.
- [25] Watanabe S. Asymptotic equivalence of bayes cross validation and widely applicable information criterion in singular learning theory. *J Mach Learn Res.* **2010** (Dec);11:3571–3594.
- [26] Rosa G, Padovani CR, Gianola D. Robust linear mixed models with normal/independent distributions and bayesian mcmc implementation. *Biometrical J: J Math Meth Biosciences.* **2003**;45(5):573–590.
- [27] Zhou J, Han L, Liu S. Nonlinear mixed-effects state space models with applications to hiv dynamics. *Stat Probab Lett.* **2013**;83(5):1448–1456.
- [28] Zhou J, Tang A, Feng H. Monte carlo likelihood estimation of mixed-effects state space models with application to hiv dynamics. *J Syst Sci and Complexity.* **2016**;29(4):1160–1176.
- [29] Shaby B, Wells MT. Exploring an adaptive metropolis algorithm. *Currently Under Rev.* 2010; 1(1):17.

Appendices

Appendix 1. Full conditionals of parameters

- Full conditional distribution of population parameter θ , given a normal prior $\mathcal{N}(\eta, \Delta)$ is expressed as:

$$\begin{aligned}
 p(\theta | \Theta, \eta, \mathbf{D}, \Delta) &\propto \exp \left\{ -\frac{1}{2} (\theta - \eta)^T \Delta^{-1} (\theta - \eta) \right\} \prod_{i=1}^m \exp \left\{ -\frac{1}{2} (\theta_i - \theta)^T \mathbf{D}^{-1} (\theta_i - \theta) \right\} \\
 &\propto \exp \left\{ -\frac{1}{2} \left[\theta^T \Delta^{-1} \theta - 2\eta^T \Delta^{-1} \theta + \sum_{i=1}^m (-2\theta_i^T \mathbf{D}^{-1} \theta + \theta^T \mathbf{D}^{-1} \theta_i) \right] \right\} \\
 &\propto \exp \left\{ -\frac{1}{2} \left[\theta^T \Delta^{-1} \theta + m\theta^T \mathbf{D}^{-1} \theta - 2 \left(\eta^T \Delta^{-1} \theta + \left(\sum_{i=1}^m \theta_i^T \right) \mathbf{D}^{-1} \theta \right) \right] \right\} \\
 &\propto \exp \left\{ -\frac{1}{2} \left[\theta^T (\Delta^{-1} + m\mathbf{D}^{-1}) \theta - 2 \left(\eta^T \Delta^{-1} + \left(\sum_{i=1}^m \theta_i^T \right) \mathbf{D}^{-1} \right) \theta \right] \right\} \\
 &\propto \exp \left\{ -\frac{1}{2} [\theta^T \mathbf{V}_1 \theta - 2\mathbf{b}_1^T \theta] \right\} \tag{A1}
 \end{aligned}$$

which is the kernel of a $\mathcal{N}(\mathbf{V}_1 \mathbf{b}_1, \mathbf{V}_1)$ density where $\mathbf{V}_1 = (\Delta^{-1} + m\mathbf{D}^{-1})^{-1}$ and $\mathbf{b}_1 = (\Delta^{-1} \eta^T + \mathbf{D}^{-1} \sum_{i=1}^m \theta_i^T)^T$.

- Full conditional distribution of $\mathbf{Q}$, given a Wishart prior $\mathbf{Q}^{-1} \sim W(\Upsilon_1^{-1}, \rho_1)$:

$$p(\mathbf{Q}^{-1}) \propto |\mathbf{Q}|^{-((\rho_1 - p - 1)/2)} \exp \left\{ -\frac{1}{2} \text{Tr}(\Upsilon_1 \mathbf{Q}^{-1}) \right\}$$

where $\mathbf{Q}^{-1}$ is a $p \times p$ matrix, is given by

$$\begin{aligned}
 p(\mathbf{Q}^{-1} | \mathbf{x}, \Theta, \Upsilon_1, \rho_1) &\propto \prod_{i=1}^m \prod_{t=1}^{n_i} |\mathbf{Q}|^{1/2} \exp \left\{ -\frac{1}{2} (\mathbf{x}_{it} - \mathbf{F}(\theta_i) \mathbf{x}_{i,t-1})^T \mathbf{Q}^{-1} (\mathbf{x}_{it} - \mathbf{F}(\theta_i) \mathbf{x}_{i,t-1}) \right\} \\
 &\times p(\mathbf{Q}^{-1})
 \end{aligned}$$

$$\propto \exp \left\{ -\frac{1}{2} \sum_{i=1}^m \sum_{t=1}^{n_i} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1})^T \mathbf{Q}^{-1} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1}) \right\} \\ \times p(\mathbf{Q}^{-1}) |\mathbf{Q}|^{-((\sum_{i=1}^m n_i)/2)}.$$

Note that

$$\sum_{i=1}^m \sum_{t=1}^{n_i} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1})^T \mathbf{Q}^{-1} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1}) = \text{Tr}(\mathbf{Q}^{-1} S_x)$$

where

$$S_x = \sum_{i=1}^m \sum_{t=1}^{n_i} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1})^T (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1}).$$

Now,

$$p(\mathbf{Q}^{-1} | \mathbf{x}, \boldsymbol{\Theta}, \boldsymbol{\Upsilon}_1, \rho_1) \propto |\mathbf{Q}|^{-(\rho_1 - p - 1)/2} \exp \left\{ -\frac{1}{2} \text{Tr}(\boldsymbol{\Upsilon}_1 \mathbf{Q}^{-1}) \right\} |\mathbf{Q}|^{((-\sum_{i=1}^m n_i)/2)} \\ \times \exp \left\{ -\frac{1}{2} \text{Tr}(S_x \mathbf{Q}^{-1}) \right\} \\ \propto |\mathbf{Q}|^{-(\sum n_i + \rho_1 - p - 1)/2} \exp \left\{ -\frac{1}{2} [\text{Tr}(\boldsymbol{\Upsilon}_1 \mathbf{Q}^{-1}) + \text{Tr}(S_x \mathbf{Q}^{-1})] \right\} \\ \propto |\mathbf{Q}|^{-(s + \rho_1 - p - 1)/2} \exp \left\{ -\frac{1}{2} \text{Tr}[(\boldsymbol{\Upsilon}_1 + S_x) \mathbf{Q}^{-1}] \right\}. \quad (\text{A2})$$

Then $\mathbf{Q}^{-1} | \mathbf{x}, \boldsymbol{\Theta}, \boldsymbol{\Upsilon}_1^{-1}, \rho_1 \sim W(\boldsymbol{\Upsilon}_1 + S_x, \rho_1 + s)$. Similarly, we can obtain full conditional posterior of matrices $\mathbf{R}$ and $\mathbf{D}$ given by

$$\mathbf{R}^{-1} | \mathbf{x}, \mathbf{y}, \boldsymbol{\Upsilon}_2^{-1}, \rho_2 \sim W(\boldsymbol{\Upsilon}_2 + S_{yx}, \rho_2 + s), \\ \mathbf{D}^{-1} | \boldsymbol{\Theta}, \boldsymbol{\theta}, \boldsymbol{\Upsilon}_3^{-1}, \rho_3 \sim W(\boldsymbol{\Upsilon}_3 + S_\theta, \rho_3 + s), \quad (\text{A3})$$

where $s = \sum_{i=1}^m n_i$, $S_{yx} = \sum_{i=1}^m \sum_{t=1}^{n_i} \lambda_{it} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it}) (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it})^T$, $S_\theta = \sum_{i=1}^m (\boldsymbol{\theta}_i - \boldsymbol{\theta})^T (\boldsymbol{\theta}_i - \boldsymbol{\theta})$, $\mathbf{V}_1 = (\boldsymbol{\Delta}^{-1} + m\mathbf{D}^{-1})^{-1}$, and $\mathbf{b}_1 = (\boldsymbol{\Delta}^{-1} \boldsymbol{\eta}^T + \mathbf{D}^{-1} \sum_{i=1}^m \boldsymbol{\theta}_i)^T$.

- Full conditional distribution of σ_Q^{-2} , given a Inverse-Gamma prior $\mathcal{G}(v_0, v_1)$ is expressed as

$$p(\sigma_Q^{-2} | \mathbf{x}, \boldsymbol{\Theta}, v_0, v_1) \propto \sigma_Q^{-2(v_0+1)} \exp \left\{ -v_1 \sigma_Q^{-2} \right\} \\ \times \prod_{i=1}^m \prod_{t=1}^{n_i} \sigma_Q^{-1} \exp \left\{ -\frac{1}{2\sigma_Q^2} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1})^T (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1}) \right\} \\ \propto \exp \left\{ -\sigma_Q^{-2} \left[v_1 + \frac{1}{2} \sum_{i=1}^m \sum_{t=1}^{n_i} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1})^T (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1}) \right] \right\} \\ \times \sigma_Q^{-2(\sum n_i + v_0 + 1)}.$$

in a similar way, we can obtain full conditional posterior of σ_R^2 and σ_D^2 when $\mathbf{R} = \sigma_R^2 \mathbf{I}$ and $\mathbf{D} = \sigma_D^2 \mathbf{I}$.

- Full conditional posterior of each θ_i is given by

$$p(\theta_i | \mathbf{y}, \mathbf{x}, \boldsymbol{\theta}, \mathbf{R}, \mathbf{Q}, \mathbf{D}, \lambda) \propto \exp \left\{ -\frac{1}{2} (\boldsymbol{\theta}_i - \boldsymbol{\theta})^T \mathbf{D}^{-1} (\boldsymbol{\theta}_i - \boldsymbol{\theta}) \right\}$$

$$\begin{aligned} & \times \prod_{t=1}^{n_i} \exp \left\{ -\frac{1}{2} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1})^T \mathbf{Q}^{-1} (\mathbf{x}_{it} - \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1}) \right\} \\ & \times \prod_{t=1}^{n_i} \exp \left\{ -\frac{\lambda_{it}}{2} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it})^T \mathbf{R}^{-1} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it}) \right\}. \quad (\text{A4}) \end{aligned}$$

Let $\mathbf{U}_1$ and $\mathbf{U}_2$ be matrices with dimensions $p^2 \times r$ and $q^2 \times r$ such as $\text{vec}(\mathbf{F}(\boldsymbol{\theta}_i)) = \mathbf{U}_1 \boldsymbol{\theta}_i$ and $\text{vec}(\mathbf{G}(\boldsymbol{\theta}_i)) = \mathbf{U}_2 \boldsymbol{\theta}_i$ respectively. We consider

$$\begin{aligned} \mathbf{F}(\boldsymbol{\theta}_i) \mathbf{x}_{i,t-1} &= (\mathbf{x}_{i,t-1}^T \otimes \mathbf{I}_p) \mathbf{U}_1 \boldsymbol{\theta}_i = \mathbf{T}_{i,t-1} \boldsymbol{\theta}_i \\ \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it} &= (\mathbf{x}_{i,t}^T \otimes \mathbf{I}_q) \mathbf{U}_2 \boldsymbol{\theta}_i = \mathbf{L}_{i,t} \boldsymbol{\theta}_i \end{aligned} \quad (\text{A5})$$

and substituting (A5) in (A4) we obtain

$$\begin{aligned} & p(\boldsymbol{\theta}_i | \mathbf{y}, \mathbf{x}, \boldsymbol{\theta}, \mathbf{R}, \mathbf{Q}, \mathbf{D}, \lambda) \\ & \propto \exp \left\{ -\frac{1}{2} \sum_{t=1}^{n_i} \lambda_{it} (\mathbf{y}_{it} - \mathbf{L}_{i,t} \boldsymbol{\theta}_i)^T \mathbf{R}^{-1} (\mathbf{y}_{it} - \mathbf{L}_{i,t} \boldsymbol{\theta}_i) \right\} \\ & \times \exp \left\{ -\frac{1}{2} (\boldsymbol{\theta}_i - \boldsymbol{\theta})^T \mathbf{D}^{-1} (\boldsymbol{\theta}_i - \boldsymbol{\theta}) - \frac{1}{2} \sum_{t=1}^{n_i} (\mathbf{x}_{it} - \mathbf{T}_{i,t-1} \boldsymbol{\theta}_i)^T \mathbf{Q}^{-1} (\mathbf{x}_{it} - \mathbf{T}_{i,t-1} \boldsymbol{\theta}_i) \right\} \\ & \propto \exp \left\{ -\frac{1}{2} [\boldsymbol{\theta}_i^T \mathbf{D}^{-1} \boldsymbol{\theta}_i - 2 \boldsymbol{\theta}^T \mathbf{D}^{-1} \boldsymbol{\theta}_i] \right\} \\ & \times \exp \left\{ -\frac{1}{2} \left[\sum_{t=1}^{n_i} \lambda_{it} (\boldsymbol{\theta}_i^T \mathbf{L}_{i,t}^T \mathbf{R}^{-1} \mathbf{L}_{i,t} \boldsymbol{\theta}_i - 2 \mathbf{y}_{it}^T \mathbf{R}^{-1} \mathbf{L}_{i,t} \boldsymbol{\theta}_i) \right] \right\} \\ & \times \exp \left\{ -\frac{1}{2} \sum_{t=1}^{n_i} (\boldsymbol{\theta}_i^T \mathbf{T}_{i,t-1}^T \mathbf{Q}^{-1} \mathbf{T}_{i,t-1} \boldsymbol{\theta}_i - 2 \mathbf{x}_{it}^T \mathbf{Q}^{-1} \mathbf{T}_{i,t-1} \boldsymbol{\theta}_i) \right\} \\ & \propto \exp \left\{ -\frac{1}{2} [\boldsymbol{\theta}_i^T \mathbf{V}_2^{-1} \boldsymbol{\theta}_i - 2 \mathbf{b}_2^T \boldsymbol{\theta}_i] \right\} \end{aligned}$$

which is the kernel of a $\mathcal{N}(\mathbf{V}_2 \mathbf{b}_2, \mathbf{V}_2)$ density where

$$\mathbf{b}_2 = \left(\boldsymbol{\theta}^T \mathbf{D}^{-1} + \sum_{t=1}^{n_i} [\lambda_{it} \mathbf{y}_{it}^T \mathbf{R}^{-1} \mathbf{L}_{i,t} + \mathbf{x}_{it}^T \mathbf{Q}^{-1} \mathbf{T}_{i,t-1}] \right)^T$$

and

$$\mathbf{V}_2 = \left(\mathbf{D}^{-1} + \sum_{t=1}^{n_i} [\lambda_{it} \mathbf{L}_{i,t}^T \mathbf{R}^{-1} \mathbf{L}_{i,t} + \mathbf{T}_{i,t-1}^T \mathbf{Q}^{-1} \mathbf{T}_{i,t-1}] \right)^{-1}.$$

- Full conditional posterior of each λ_{it} is expressed as:

$$\begin{aligned} p(\lambda_{it} | \mathbf{x}_{it}, \mathbf{y}_{it}, \mathbf{R}, v_i) & \propto \lambda_{it}^{1/2} \exp \left\{ -\frac{\lambda_{it}}{2} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it})^T \mathbf{R}^{-1} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it}) \right\} \lambda_{it}^{(v_i/2)-1} \\ & \times \exp \left\{ -\frac{v_i}{2} \lambda_{it} \right\} \\ & \propto \lambda_{it}^{((v_i+1)/2)-1} \exp \left\{ -\frac{\lambda_{it}}{2} [(\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it})^T \mathbf{R}^{-1} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it}) + v_i] \right\} \end{aligned}$$

which is the kernel of a $\mathcal{G}(\alpha_i, \beta_i)$ density where $\alpha_i = (v_i + 1)/2$ and

$$\beta_i = \frac{1}{2} \left[v_i + (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it})^T \mathbf{R}^{-1} (\mathbf{y}_{it} - \mathbf{G}(\boldsymbol{\theta}_i) \mathbf{x}_{it}) \right]$$

- Metropolis-Hastings algorithm for each v_i .

Consider the following log posterior distribution for each v_i ,

$$\ell(v_i | \lambda_i) \propto \log(p(v_i)) + \frac{v_i n_i}{2} \log\left(\frac{v_i}{2}\right) - n_i \log\left(\Gamma\left(\frac{v_i}{2}\right)\right) - v_i \left[\frac{1}{2} \sum_{t=1}^{n_i} (\log(\lambda_{it}) + \lambda_{it}) \right], \quad (\text{A6})$$

where $p(v_i)$ is the respective prior. In order to sample from (A6) we used an adaptive Metropolis algorithm based on [29] given by

Algorithm A.1: 1) Set $j = 0$ and get starting values for $v_i^{(0)}$ and the quantities $m_0^{(0)}$, $s_0^{(0)}$ and $\sigma_0^{(0)}$.

2) For each iteration (j)

2.1) $gg = 1/j^{0.8}$.

2.2) Generate $e \sim \mathcal{N}(0, 1)$ truncated in $[4, 40]$ and compute $v_p = v_i^{(j-1)} + e \sqrt{s_0^{(j-1)} \sigma_0^{(j-1)}}$.

2.3) Compute $\alpha = \min\{1, \exp[\ell(v_p | \lambda_i^{(j)}) - \ell(v_i^{(j-1)} | \lambda_i^{(j)})]\}$ and generate $u \sim \mathcal{U}(0, 1)$; if $u < \alpha$ then $v_i^{(j)} = v_p$, else $v_i^{(j)} = v_i^{(j-1)}$

2.4) Update the quantities:

$$\begin{aligned} s_0^{(j)} &= \exp\{\log(s_0^{(j-1)}) + gg(\alpha - 0.234)\} \\ \sigma_0^{(j)} &= \sigma_0^{(j-1)} + gg[(v_i^{(j)} - m_0^{(j-1)})^2 - \sigma_0^{(j-1)}] \\ m_0^{(j)} &= m_0^{(j-1)} + gg(v_i^{(j)} - m_0^{(j-1)}). \end{aligned}$$

Appendix 2. Details of algorithm 2.2

Consider the following general linear state-space model

$$\begin{aligned} \mathbf{y}_t &= \mathbf{J}_t \boldsymbol{\beta} + \mathbf{Z}_t \mathbf{x}_t + \mathbf{B}_t \mathbf{u}_t \quad t = 1, 2, \dots, n \\ \mathbf{x}_{t+1} &= \mathbf{W}_t \boldsymbol{\beta} + \mathbf{T}_t \mathbf{x}_t + \mathbf{K}_t \mathbf{u}_t, \quad t = 1, 2, \dots, n-1, \\ \mathbf{x}_1 &\sim \mathcal{N}_p(\mathbf{a}_1, \mathbf{P}_1), \quad \mathbf{u}_t \sim \mathcal{N}_{p+q}(\mathbf{0}, \mathbf{I}). \end{aligned}$$

It is straightforward to see that the MESSM_t model (for a single individual) is an special case of this general model when $\mathbf{J}_t = \mathbf{W}_t = \mathbf{0}$, $\mathbf{Z}_t = \mathbf{G}_t$, $\mathbf{B}_t \mathbf{B}_t^T = \mathbf{R} \lambda_t^{-1}$, $\mathbf{T}_t = \mathbf{F}_t$, $\mathbf{K}_t \mathbf{K}_t^T = \mathbf{Q}$ and $\mathbf{u}_t = (\mathbf{v}_t, \mathbf{w}_t)^T$,

$$\begin{aligned} \mathbf{y}_t &= \mathbf{G}_t \mathbf{x}_t + \mathbf{v}_t \quad t = 1, 2, \dots, n \\ \mathbf{x}_t &= \mathbf{F}_t \mathbf{x}_t + \mathbf{w}_t, \quad t = 2, \dots, n, \\ \mathbf{x}_1 | \mathbf{x}_0 &\sim \mathcal{N}(\mathbf{a}_1 = \mathbf{F}_1 \mathbf{x}_0, \mathbf{P}_1 = \mathbf{Q}) \\ \mathbf{x}_0 &\sim \mathcal{N}_p(\boldsymbol{\tau}, \mathbf{N}) \\ \mathbf{u}_t &\sim \mathcal{N}_{p+q}\left(\mathbf{0}, \begin{bmatrix} \mathbf{R} \lambda_t^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} \end{bmatrix}\right). \end{aligned}$$

- First step in using MMP method is to define the matrix $\mathbf{A}_t$, as the precision matrix of the stacked period-t innovation $\mathbf{u}_t$,

$$\mathbf{A}_t = \begin{bmatrix} \mathbf{R} \lambda_t^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} \end{bmatrix}^{-1} = \begin{bmatrix} \lambda_t \mathbf{R}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q}^{-1} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11,t} & \mathbf{A}_{12,t} \\ \mathbf{A}_{21,t} & \mathbf{A}_{22,t} \end{bmatrix}.$$

Then, we compute de precision matrix Ω and co-vector c of the conditional distribution of the states $\mathbf{x}$ given the observations $\mathbf{y}$ following the expressions in Appendix A of [16]. Therefore we have

$$\begin{aligned}\Omega_{11} &= \mathbf{G}_1^T \mathbf{A}_{11,1} \mathbf{G}_1 + \mathbf{G}_1^T \mathbf{A}_{12,1} \mathbf{F}_1 + \mathbf{F}_1^T \mathbf{A}_{21,1} \mathbf{G}_1 + \mathbf{F}_1^T \mathbf{A}_{22,1} \mathbf{F}_1 + \mathbf{P}^{-1} \\ &= \lambda_t \mathbf{R}^{-1} + \mathbf{F}_1^T \mathbf{Q}^{-1} \mathbf{F}_1 + \mathbf{Q}^{-1} \\ \Omega_{tt} &= \mathbf{G}_t^T \mathbf{A}_{11,t} \mathbf{G}_t + \mathbf{G}_t^T \mathbf{A}_{12,t} \mathbf{F}_t + \mathbf{F}_t^T \mathbf{A}_{21,t} \mathbf{G}_t + \mathbf{F}_t^T \mathbf{A}_{22,t} \mathbf{F}_t + \mathbf{A}_{22,t-1} \\ &= \lambda_t \mathbf{R}^{-1} + \mathbf{F}_t^T \mathbf{Q}^{-1} \mathbf{F}_t + \mathbf{Q}^{-1} \\ \Omega_{t,t-1} &= -\mathbf{G}_t^T \mathbf{A}_{12,t} - \mathbf{F}_t^T \mathbf{A}_{22,t} = -\mathbf{F}_t^T \mathbf{Q}^{-1} \\ c_1 &= \mathbf{G}_1^T \mathbf{A}_{11,1} \mathbf{y}_1 + \mathbf{P}^{-1} \mathbf{a}_1 = \lambda_t \mathbf{R}^{-1} \mathbf{y}_1 + \mathbf{Q}^{-1} \mathbf{F}_1 \mathbf{x}_0 \\ c_t &= \mathbf{G}_t^T \mathbf{A}_{11,t} \mathbf{y}_t = \lambda_t \mathbf{R}^{-1} \mathbf{y}_t.\end{aligned}$$

Finally, for $t = n-1, n-2, \dots, 1$, we sample from $\mathbf{x}_t | \mathbf{x}_{t+1}, \dots, \mathbf{x}_n, \mathbf{y} \sim \mathcal{N}(m_t - \Sigma_t \Omega_{t,t+1} \mathbf{x}_{t+1}, \Sigma_t)$, where

$$\begin{aligned}\Sigma_1 &= (\Omega_{11})^{-1}, \quad m_1 = \Sigma_1 c_1, \\ \Sigma_t &= (\Omega_{tt} - \Omega_{t-1,t}^T \Sigma_{t-1} \Omega_{t-1,t})^{-1}, \quad m_t = \Sigma_t (c_t - \Omega_{t-1,t}^T m_{t-1}).\end{aligned}$$

For $t = n$, we sample from $\mathbf{x}_n | \mathbf{y} \sim \mathcal{N}(m_n, \Sigma_n)$.

- The value of x_0 , that is used in the calculation of the co-vector within the algorithm, is sampled from the posteriori distribution

$$\begin{aligned}p(\mathbf{x}_0 | \mathbf{x}_1) &\propto \exp \left\{ -\frac{1}{2} (\mathbf{x}_1 - \mathbf{F}_1 \mathbf{x}_0)^T \mathbf{Q}^{-1} (\mathbf{x}_1 - \mathbf{F}_1 \mathbf{x}_0) \right\} \exp \left\{ -\frac{1}{2} (\mathbf{x}_0 - \boldsymbol{\tau})^T \mathbf{N}^{-1} (\mathbf{x}_0 - \boldsymbol{\tau}) \right\} \\ &\propto \exp \left\{ -\frac{1}{2} \left[\mathbf{x}_0^T \mathbf{F}_1^T \mathbf{Q}^{-1} \mathbf{F}_1 \mathbf{x}_0 - 2 \mathbf{x}_1^T \mathbf{F}_1 \mathbf{x}_0 + \mathbf{x}_0^T \mathbf{N}^{-1} \mathbf{x}_0 - 2 \boldsymbol{\tau}^T \mathbf{x}_0 \right] \right\} \\ &\propto \exp \left\{ -\frac{1}{2} \left[\mathbf{x}_0^T \left(\mathbf{F}_1^T \mathbf{Q}^{-1} \mathbf{F}_1 + \mathbf{N}^{-1} \right) \mathbf{x}_0 - 2 \left(\mathbf{x}_1^T \mathbf{F}_1 + \boldsymbol{\tau}^T \right) \mathbf{x}_0 \right] \right\}\end{aligned}$$

which is the kernel of a $\mathcal{N}(\mathbf{V}_3 \mathbf{b}_3, \mathbf{V}_3)$ density where $\mathbf{V}_3 = (\mathbf{F}_1^T \mathbf{Q}^{-1} \mathbf{F}_1 + \mathbf{N}^{-1})^{-1}$ and $\mathbf{b}_3 = (\mathbf{x}_1^T \mathbf{F}_1 + \boldsymbol{\tau}^T)^T$.